

Exercise 3E Solutions Worksheet

Terminating, Recurring, and Vulgar Fractions Solutions Guide

Textbook Working Rules for Recurring Decimals:

- **Recurring/Repeating Decimal:** A decimal in which a digit or set of digits repeats continuously. To write it in short form, put a bar (e.g., $0.\overline{6}$) over the repeating digits. In parenthesis notation, this is written as $0.(6)$.
- **Pure Recurring Decimal to Vulgar Fraction:** Write the repeating digits in the numerator and place as many 9s in the denominator as there are repeating digits (e.g., $0.\overline{34} = 34/99$).
- **Mixed Recurring Decimal to Vulgar Fraction:** In the numerator, subtract the non-repeating digits from the entire decimal part. In the denominator, write as many 9s as there are repeating digits followed by as many zeros as there are non-repeating digits (e.g., $0.2\overline{1} = (21 - 2)/90 = 19/90$).

Question 1: Express each of the following as a recurring decimal:

To express a fraction as a recurring decimal, we carry out long division. Once the remainder starts repeating, we identify the repeating cycle of digits and write a bar over them.

Part	Fraction	Long Division Steps & Cycle	Recurring Decimal Representation
(i)	$20/3$	Dividing 20 by 3: $20 = 3 \times 6 + 2$ (Remainder 2) Bring down 0 to make 20, $20 = 3 \times 6 + 2$ (Remainder 2) The quotient is 6.6666... and remainder 2 repeats continuously.	6.6 (or $6.(6)$)
(ii)	$3/11$	Dividing 3 by 11: $3.0 \div 11 = 0.2$ (Remainder 8) $80 \div 11 = 7$ (Remainder 3) $30 \div 11 = 2$ (Remainder 8) The block '27' repeats continuously.	0.27 (or $0.(27)$)
(iii)	$5/6$	Dividing 5 by 6: $5.0 \div 6 = 0.8$ (Remainder 2) $20 \div 6 = 3$ (Remainder 2) $20 \div 6 = 3$ (Remainder 2) The digit '3' repeats continuously after '8'.	0.83 (or $0.8(3)$)

Part	Fraction	Long Division Steps & Cycle	Recurring Decimal Representation
(iv)	17/90	Dividing 17 by 90: $17.0 \div 90 = 0.1$ (Remainder 80) $800 \div 90 = 8$ (Remainder 80) $800 \div 90 = 8$ (Remainder 80) The digit '8' repeats continuously after '1'.	0.18 (or 0.1(8))
(v)	1/37	Dividing 1 by 37: $1.00 \div 37 = 0.02$ (Remainder 26) $260 \div 37 = 7$ (Remainder 1) $100 \div 37 = 2$ (Remainder 26) The block '027' repeats continuously.	0.027 (or 0.(027))
(vi)	22/7	Dividing 22 by 7: $22 \div 7 = 3$ (Remainder 1) $10 \div 7 = 1$ (Remainder 3) $30 \div 7 = 4$ (Remainder 2) $20 \div 7 = 2$ (Remainder 6) $60 \div 7 = 8$ (Remainder 4) $40 \div 7 = 5$ (Remainder 5) $50 \div 7 = 7$ (Remainder 1) The block '142857' repeats continuously.	3.142857 (or 3.(142857))
(vii)	2/13	Dividing 2 by 13: $2.0 \div 13 = 0.1$ (Remainder 7) $70 \div 13 = 5$ (Remainder 5) $50 \div 13 = 3$ (Remainder 11) $110 \div 13 = 8$ (Remainder 6) $60 \div 13 = 4$ (Remainder 8) $80 \div 13 = 6$ (Remainder 2) $20 \div 13 = 1$ (Remainder 7) The block '153846' repeats continuously.	0.153846 (or 0.(153846))

Question 2: Convert each of the following into a vulgar fraction:

Rule for Pure Recurring Decimals: Write the repeating digits as the numerator, and use as many 9s in the denominator as there are repeating digits. Simplify if possible.

Part	Recurring Decimal	Mathematical Conversion Steps	Vulgar Fraction (Simplest Form)
(i)	$0.\overline{6}$	- Repeating digit is 6 (1 digit repeating). - Numerator = 6 - Denominator = 9 - Fraction = $6/9 = (6 \div 3)/(9 \div 3)$	$2/3$
(ii)	$0.\overline{8}$	- Repeating digit is 8 (1 digit repeating). - Numerator = 8 - Denominator = 9	$8/9$
(iii)	$0.\overline{34}$	- Repeating digits are 34 (2 digits repeating). - Numerator = 34 - Denominator = 99	$34/99$
(iv)	$2.\overline{13}$ (or $2.1\overline{3}$)	Let $x = 2.1313\ldots$ (bar on 13) $100x = 213.1313\ldots$ - Subtracting: $99x = 211$ $x = 211/99 = 2\frac{13}{99}$. Alternative (if bar is only on 3, i.e. $2.1\overline{3}$): Let $x = 2.1333\ldots$ $10x = 21.333\ldots$ and $100x = 213.333\ldots$ - Subtracting: $90x = 192 \rightarrow x = 192/90 = 32/15 = 2\frac{2}{15}$.	$2\frac{13}{99}$ (or $2\frac{2}{15}$)
(v)	$1.\overline{243}$	Let $x = 1.243243\ldots$ (bar on 243) $1000x = 1243.243243\ldots$ - Subtracting: $999x = 1242$ $x = 1242/999 = (1242 \div 27)/(999 \div 27) = 46/37 = 1\frac{9}{37}$.	$1\frac{9}{37}$ (or $46/37$)

Question 3: Convert each of the following into a vulgar fraction:

Rule for Mixed Recurring Decimals:

- Numerator:** Take the difference between the full number formed by all decimal digits and the number formed by the non-repeating digits.
- Denominator:** Write as many 9s as there are repeating digits, followed by as many zeros as there are non-repeating digits.

Part	Mixed Recurring	Conversion Details & Steps	Vulgar Fraction (Simplest Form)
(i)	$0.1\overline{6}$ (bar on 6)	- Repeating part = 6 (one digit), Non-repeating part = 1 (one digit). - Numerator = $16 - 1 = 15$ - Denominator = one 9 followed by one 0 = 90 - Fraction = $15/90 = (15 \div 15)/(90 \div 15)$	1/6
(ii)	$0.1\overline{43}$ (bar on 43)	- Repeating part = 43 (two digits), Non-repeating part = 1 (one digit). - Numerator = $143 - 1 = 142$ - Denominator = two 9s followed by one 0 = 990 - Fraction = $142/990 = 71/495$. Alternative (if bar is only on 3, i.e. $0.14\overline{3}$): - Numerator = $143 - 14 = 129$ - Denominator = one 9 followed by two 0s = 900 - Fraction = $129/900 = 43/300$.	71/495 (or 43/300)
(iii)	$0.57\overline{4}$ (bar on 4)	- Repeating part = 4 (one digit), Non-repeating part = 57 (two digits). - Numerator = $574 - 57 = 517$ - Denominator = one 9 followed by two 0s = 900 - Fraction = $517/900$. Alternative (if bar is on 74, i.e. $0.57\overline{4}$): - Numerator = $574 - 5 = 569$ - Denominator = two 9s followed by one 0 = 990 - Fraction = $569/990$.	517/900 (or 569/990)
(iv)	$0.123\overline{4}$ (bar on 34)	- Repeating part = 34 (two digits), Non-repeating part = 12 (two digits). - Numerator = $1234 - 12 = 1222$ - Denominator = two 9s followed by two 0s = 9900 - Fraction = $1222/9900 = 611/4950$. Alternative (if bar is only on 4, i.e. $0.123\overline{4}$): - Numerator = $1234 - 123 = 1111$ - Denominator = one 9 followed by three 0s = 9000 - Fraction = $1111/9000$.	611/4950 (or 1111/9000)